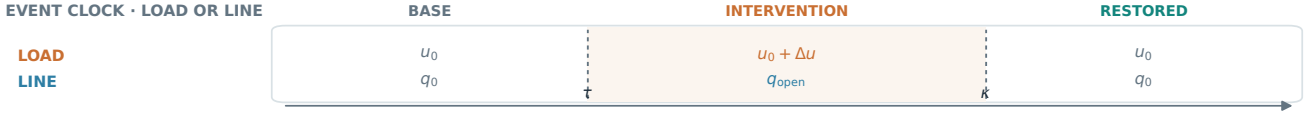
FROM EVENT TO CAUSAL PMU REPORT

The intervention ends at OFF; its physical and measurement response continues.



SWITCHED PLANT

$$\dot{x} = f_q(x, z, u)$$
$$0 = g_q(x, z, u)$$

differential state and network equations

ELECTRICAL MAP

$$h_k = H_q(x, z, u)$$

bus-voltage phasors · oriented terminal currents

CAUSAL PMU REPORT

$$\eta_k = F_M(\eta_{k-1}, h_k)$$
$$y_k \in \mathbb{R}^{47}$$

phasor, frequency and ROCOF channels

RESTORATION RETURNS THE OPERATOR, NOT THE RESPONSE

u, q return to base · x, η persist · z and terminal I are recomputed

At report k, inference receives $y_{1:k}$; event truth and raw waveforms are not inputs.